

all acts detrimental to the preservation of peaceful social cooperation and to the improvement of interhuman relations. They welcome the support which religious teachings may give to those moral precepts of which they themselves approve, but they are opposed to all those norms which are bound to bring about social disintegration from whatever source they may stem.

It is a distortion of fact to say, as many champions of religious theocracy do, that liberalism fights religion. Where the principle of church interference with secular issues is in force, the various churches, denominations and sects are fighting one another. By separating church and state, liberalism establishes peace between the various religious factions and gives to each of them the opportunity to preach its gospel unmolested.

Liberalism is rationalistic. It maintains that it is possible to convince the immense majority that peaceful cooperation within the framework of society better serves their rightly understood interests than mutual battling and social disintegration. It has full confidence in man's reason. It may be that this optimism is unfounded and that the liberals have erred. But then there is no hope left for mankind's future.

3. The Division of Labor

The fundamental social phenomenon is the division of labor and its counterpart human cooperation.

Experience teaches man that cooperative action is more efficient and productive than isolated action of self-sufficient individuals. The natural conditions determining man's life and effort are such that the division of labor increases output per unit of labor expended. These natural facts are:

First: the innate inequality of men with regard to their ability to perform various kinds of labor. Second: the unequal distribution of the nature-given, nonhuman opportunities of production on the surface of the earth. One may as well consider these two facts as one and the same fact, namely, the manifoldness of nature which makes the universe a complex of infinite varieties. If the earth's surface were such that the physical conditions of production were the same at every point and if one man were as equal to all other men as is a circle to another with the same diameter in Euclidian geometry, division of labor would not offer any advantages for acting man.

There is still a third fact, viz., that there are undertakings whose accomplishment exceeds the forces of a single man and requires the joint effort of several. Some of them require an expenditure of labor which no single man can perform because his capacity to work is not great enough. Others again could be accomplished by individuals;

but the time which they would have to devote to the work would be so long that the result would only be attained late and would not compensate for the labor expended. In both cases only joint effort makes it possible to attain the end sought.

If only this third condition were present, temporary cooperation between men would have certainly emerged. However, such transient alliances to cope with specific tasks which are beyond the strength of an individual would not have brought about lasting social cooperation. Undertakings which could be performed only in this way were not very numerous at the early stages of civilization. Moreover, all those concerned may not often agree that the performance in question is more useful and urgent than the accomplishment of other tasks which they could perform alone. The great human society enclosing all men in all of their activities did not originate from such occasional alliances. Society is much more than a passing alliance concluded for a definite purpose and ceasing as soon as its objective is realized, even if the partners are ready to renew it should an occasion present itself.

The increase in productivity brought about by the division of labor is obvious whenever the inequality of the participants is such that every individual or every piece of land is superior at least in one regard to the other individuals or pieces of land concerned. If A is fit to produce in 1 unit of time $6p$ or $4q$ and B only $2p$, but $8q$, they both, when working in isolation, will produce together $4p + 6q$; when working under the division of labor, each of them producing only that commodity in whose production he is more efficient than his partner, they will produce $6p + 8q$. But what will happen, if A is more efficient than B not only in the production of p but also in the production of q ?

This is the problem which Ricardo raised and solved immediately.

4. The Ricardian Law of Association

Ricardo expounded the law of association in order to demonstrate what the consequences of the division of labor are when an individual or a group, more efficient in every regard, cooperates with an individual or a group less efficient in every regard. He investigated the effects of trade between two areas, unequally endowed by nature, under the assumption that the products, but not the workers and the accumulated factors of future production (capital goods), can freely move from each area into the other. The division of labor between two such areas will, as Ricardo's law shows, increase the productivity of

labor and is therefore advantageous to all concerned, even if the physical conditions of production for any commodity are more favorable in one of these two areas than in the other. It is advantageous for the better endowed area to concentrate its efforts upon the production of those commodities for which its superiority is greater, and to leave to the less endowed area the production of other goods in which its own superiority is less. The paradox that it is more advantageous to leave more favorable domestic conditions of production unused and to procure the commodities they could produce from areas in which conditions for their production are less favorable, is the outcome of the immobility of labor and capital, to which the more favorable places of production are inaccessible.

Ricardo was fully aware of the fact that his law of comparative cost, which he expounded mainly in order to deal with a special problem of international trade, is a particular instance of the more universal law of association.

If *A* is in such a way more efficient than *B* that he needs for the production of 1 unit of the commodity *p* 3 hours compared with *B*'s 5, and for the production of 1 unit of *q* 2 hours compared with *B*'s 4, then both will gain if *A* confines himself to producing *q* and leaves *B* to produce *p*. If each of them gives 60 hours to producing *p* and 60 hours to producing *q*, the result of *A*'s labor is 20 *p* + 30 *q*; of *B*'s, 12 *p* + 15 *q*; and for both together, 32 *p* + 45 *q*. If, however, *A* confines himself to producing *q* alone, he produces 60 *q* in 120 hours, while *B*, if he confines himself to producing *p*, produces in the same time 24 *p*. The result of their activities is then 24 *p* + 60 *q*, which, as *p* has for *A* a substitution ratio of $\frac{3}{2} q$ and for *B* one of $\frac{5}{4} q$, signifies a larger output than 32 *p* + 45 *q*. Therefore it is manifest that the division of labor brings advantages to all who take part in it. Collaboration of the more talented, more able, and more industrious with the less talented, less able, and less industrious results in benefit for both. The gains derived from the division of labor are always mutual.

The law of association makes us comprehend the tendencies which resulted in the progressive intensification of human cooperation. We conceive what incentive induced people not to consider themselves simply as rivals in a struggle for the appropriation of the limited supply of means of subsistence made available by nature. We realize what has impelled them and permanently impels them to consort with one another for the sake of cooperation. Every step forward on the way to a more developed mode of the division of labor serves the interests of all participants. In order to comprehend why man did not

remain solitary, searching like the animals for food and shelter for himself only and at most also for his consort and his helpless infants, we do not need to have recourse to a miraculous interference of the Deity or to the empty hypostasis of an innate urge toward association. Neither are we forced to assume that the isolated individuals or primitive hordes one day pledged themselves by a contract to establish social bonds. The factor that brought about primitive society and daily works toward its progressive intensification is human action that is animated by the insight into the higher productivity of labor achieved under the division of labor.

Neither history nor ethnology nor any other branch of knowledge can provide a description of the evolution which has led from the packs and flocks of mankind's nonhuman ancestors to the primitive, yet already highly differentiated, societal groups about which information is provided in excavations, in the most ancient documents of history, and in the reports of explorers and travelers who have met savage tribes. The task with which science is faced in respect of the origins of society can only consist in the demonstration of those factors which can and must result in association and its progressive intensification. Praxeology solves the problem. If and as far as labor under the division of labor is more productive than isolated labor, and if and as far as man is able to realize this fact, human action itself tends toward cooperation and association; man becomes a social being not in sacrificing his own concerns for the sake of a mythical Moloch, society, but in aiming at an improvement in his own welfare. Experience teaches that this condition—higher productivity achieved under the division of labor—is present because its cause—the inborn inequality of men and the inequality in the geographical distribution of the natural factors of production—is real. Thus we are in a position to comprehend the course of social evolution.

Current Errors Concerning the Law of Association

People cavil much about Ricardo's law of association, better known under the name *law of comparative cost*. The reason is obvious. This law is an offense to all those eager to justify protection and national economic isolation from any point of view other than the selfish interests of some producers or the issues of war-preparedness.

Ricardo's first aim in expounding this law was to refute an objection raised against freedom of international trade. The protectionist asks: What under free trade will be the fate of a country in which the conditions for any kind of production are less favorable than in all other countries? Now, in a world in which there is free mobility not only for products, but no

less for capital goods and for labor, a country so little suited for production would cease to be used as the seat of any human industry. If people fare better without exploiting the—comparatively unsatisfactory—physical conditions of production offered by this country, they will not settle here and will leave it as uninhabited as the polar regions, the tundras and the deserts. But Ricardo deals with a world whose conditions are determined by settlement in earlier days, a world in which capital goods and labor are bound to the soil by definite institutions. In such a milieu free trade, i.e., the free mobility of commodities only, cannot bring about a state of affairs in which capital and labor are distributed on the surface of the earth according to the better or poorer physical opportunities afforded to the productivity of labor. Here the law of comparative cost comes into operation. Each country turns toward those branches of production for which its conditions offer comparatively, although not absolutely, the most favorable opportunities. For the inhabitants of a country it is more advantageous to abstain from the exploitation of some opportunities which—absolutely and technologically—are more propitious and to import commodities produced abroad under conditions which—absolutely and technologically—are less favorable than the unused domestic resources. The case is analogous to that of a surgeon who finds it convenient to employ for the cleaning of the operating-room and the instruments a man whom he excels in this performance also and to devote himself exclusively to surgery, in which his superiority is higher.

The theorem of comparative cost is in no way connected with the value theory of classical economics. It does not deal with value or with prices. It is an analytic judgment; the conclusion is implied in the two propositions that the technically movable factors of production differ with regard to their productivity in various places and are institutionally restricted in their mobility. The theorem, without prejudice to the correctness of its conclusions, can disregard problems of valuation because it is free to resort to a set of simple assumptions. These are: that only two products are to be produced; that these products are freely movable; that for the production of each of them two factors are required; that one of these factors (it may be either labor or capital goods) is identical in the production of both, while the other factor (a specific property of the soil) is different for each of the two processes; that the greater scarcity of the factor common to both processes determines the extent of the exploitation of the different factor. In the frame of these assumptions, which make it possible to establish substitution ratios between the expenditure of the common factor and the output, the theorem answers the question raised.

The law of comparative cost is as independent of the classical theory of value as is the law of returns, which its reasoning resembles. In both cases we can content ourselves with comparing only physical input and physical output. With the law of returns we compare the output of the same product. With the law of comparative costs we compare the output of two different products. Such a comparison is feasible because we assume that

for the production of each of them, apart from one specific factor, only nonspecific factors of the same kind are required.

Some critics blame the law of comparative cost for this simplification of assumptions. They believe that the modern theory of value would require a reformulation of the law in conformity with the principles of subjective value. Only such a formulation could provide a satisfactory conclusive demonstration. However, they do not want to calculate in terms of money. They prefer to resort to those methods of utility analysis which they consider a means for making value calculations in terms of utility. It will be shown in the further progress of our investigation that these attempts to eliminate monetary terms from economic calculation are delusive. Their fundamental assumptions are untenable and contradictory and all formulas derived from them are vicious. No method of economic calculation is possible other than one based on money prices as determined by the market.⁷

The meaning of the simple assumptions underlying the law of comparative cost is not precisely the same for the modern economists as it was for the classical economists. Some adherents of the classical school considered them as the starting point of a theory of value in international trade. We know now that they were mistaken in this belief. Besides, we realize that with regard to the determination of value and of prices there is no difference between domestic and foreign trade. What makes people distinguish between the home market and markets abroad is only a difference in the data, i.e., varying institutional conditions restricting the mobility of factors of production and of products.

If we do not want to deal with the law of comparative cost under the simplified assumptions applied by Ricardo, we must openly employ money calculation. We must not fall prey to the illusion that a comparison between the expenditure of factors of production of various kinds and of the output of products of various kinds can be achieved without the aid of money calculation. If we consider the case of the surgeon and his handyman we must say: If the surgeon can employ his limited working time for the performance of operations for which he is compensated at \$50 per hour, it is to his interest to employ a handyman to keep his instruments in good order and to pay him \$2 per hour, although this man needs 3 hours to accomplish what the surgeon could do in 1 hour. In comparing the conditions of two countries we must say: If conditions are such that in England the production of 1 unit of each of the two commodities *a* and *b* requires the expenditure of 1 working day of the same kind of labor, while in India with the same investment of capital for *a* 2 days and for *b* 3 days are required, and if capital goods and *a* and *b* are freely movable from England to India and vice versa, while there is no mobility of labor, wage rates in India in the production of *a* must tend to be 50 per cent, and in the production of *b* 33 $\frac{1}{3}$ per cent, of the English rates. If the English rate is 6 shillings, the rates in India would be the equivalent of 3 shillings in the production

7. See below, pp. 202-210.

of *a* and the equivalent of 2 shillings in the production of *b*. Such a discrepancy in the remuneration of labor of the same kind cannot last if there is mobility of labor on the domestic Indian labor market. Workers would shift from the production of *b* into the production of *a*; their migration would tend to lower the remuneration in the *a* industry and to raise it in the *b* industry. Finally Indian wage rates would be equal in both industries. The production of *a* would tend to expand and to supplant English competition. On the other hand the production of *b* would become unprofitable in India and would have to be discontinued, while it would expand in England. The same reasoning is valid if we assume that the difference in the conditions of production consists also or exclusively in the amount of capital investment needed.

It has been asserted that Ricardo's law was valid only for his age and is of no avail for our time which offers other conditions. Ricardo saw the difference between domestic trade and foreign trade in differences in the mobility of capital and labor. If one assumes that capital, labor, and products are movable, then there exists a difference between regional and interregional trade only as far as the cost of transportation comes into play. Then it is superfluous to develop a theory of international trade as distinguished from national trade. Capital and labor are distributed on the earth's surface according to the better or poorer conditions which the various regions offer to production. There are areas more densely populated and better equipped with capital, there are others less densely populated and poorer in capital supply. There prevails on the whole earth a tendency toward an equalization of wage rates for the same kind of labor.

Ricardo, however, starts from the assumption that there is mobility of capital and labor only within each country, and not between the various countries. He raises the question what the consequences of the free mobility of products must be under such conditions. (If there is no mobility of products either, then every country is economically isolated and autarkic, and there is no international trade at all.) The theory of comparative cost answers this question. Now, Ricardo's assumptions by and large held good for his age. Later, in the course of the nineteenth century, conditions changed. The immobility of capital and labor gave way; international transfer of capital and labor became more and more common. Then came a reaction. Today capital and labor are again restricted in their mobility. Reality again corresponds to the Ricardian assumptions.

However, the teachings of the classical theory of interregional trade are above any change in institutional conditions. They enable us to study the problems involved under any imaginable assumptions.

5. The Effects of the Division of Labor

The division of labor is the outcome of man's conscious reaction to the multiplicity of natural conditions. On the other hand it is itself a factor bringing about differentiation. It assigns to the various geo-

graphic areas specific functions in the complex of the processes of production. It makes some areas urban, others rural; it locates the various branches of manufacturing, mining, and agriculture in different places. Still more important, however, is the fact that it intensifies the innate inequality of men. Exercise and practice of specific tasks adjust individuals better to the requirements of their performance; men develop some of their inborn faculties and stunt the development of others. Vocational types emerge, people become specialists.

The division of labor splits the various processes of production into minute tasks, many of which can be performed by mechanical devices. It is this fact that made the use of machinery possible and brought about the amazing improvements in technical methods of production. Mechanization is the fruit of the division of labor, its most beneficial achievement, not its motive and fountain spring. Power-driven specialized machinery could be employed only in a social environment under the division of labor. Every step forward on the road toward the use of more specialized, more refined, and more productive machines requires a further specialization of tasks.

6. The Individual Within Society

If praxeology speaks of the solitary individual, acting on his own behalf only and independent of fellow men, it does so for the sake of a better comprehension of the problems of social cooperation. We do not assert that such isolated autarkic human beings have ever lived and that the social stage of man's history was preceded by an age of independent individuals roaming like animals in search of food. The biological humanization of man's nonhuman ancestors and the emergence of the primitive social bonds were effected in the same process. Man appeared on the scene of earthly events as a social being. The isolated asocial man is a fictitious construction.

Seen from the point of view of the individual, society is the great means for the attainment of all his ends. The preservation of society is an essential condition of any plans an individual may want to realize by any action whatever. Even the refractory delinquent who fails to adjust his conduct to the requirements of life within the societal system of cooperation does not want to miss any of the advantages derived from the division of labor. He does not consciously aim at the destruction of society. He wants to lay his hands on a greater portion of the jointly produced wealth than the social order assigns to him. He would feel miserable if antisocial behavior were to become universal and its inevitable outcome, the return to primitive indigence, resulted.